Oxygen-Centered Representations Reveal a Candidate Dynamical Organization of Chemical Space.

A minimal ERC-based analysis of recoverability, vibrational structure, and anchor dependence

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1. Abstract

We explore whether oxygen-centered chemical representations may reveal a hidden dynamical organization of the periodic table.

Using a minimal Endogenous Reachability Collapse (ERC) framework, we construct a reduced embedding of chemical space and evaluate whether a dominant dynamical axis emerges from the data.

Across multiple analyses, evidence consistent with a non-random, structured organization is observed, although the strength of the signal varies across tests. The first principal component organizes elements along a candidate dynamical accessibility axis, ERC-derived vulnerability varies systematically along this axis, and unsupervised clustering identifies coarse regime-like partitions.

The model is not intended to reproduce molecular spectroscopy quantitatively. Comparison with known homonuclear diatomic vibrational frequencies shows only partial and preliminary consistency with physical observables, limited by the small number of available diatomic systems.

Importantly, the weak correlation between PC1 and simple oxygen-distance metrics suggests that the observed structure is not trivially explained by geometric proximity alone.

These results are consistent with the possibility that oxygen-centered representations may expose dynamical accessibility regimes across chemical space. This remains a working hypothesis requiring further validation under alternative embeddings, extended datasets, and physically grounded models.

2. Introduction

The periodic table is typically understood as a static organization of chemical elements, structured by electronic configuration and recurring trends in observable properties.

However, many chemically and biologically relevant processes are inherently dynamical. In these systems, the key question may not be which states exist, but whether transitions between states remain dynamically accessible under perturbation.

This distinction can be framed in terms of accessibility versus admissibility. A system may retain its underlying structure while losing the ability to return to prior states, due to constraints on recovery timescales or pathway availability.

In this context, oxygen provides a natural reference point. As a central participant in redox processes across chemical and biological systems, oxygen-dependent dynamics often operate near recovery limits, where timing and accessibility may influence system behavior.

This motivates the following question: whether chemical space, when represented relative to oxygen, may exhibit an underlying dynamical organization not readily visible through static descriptors alone.

Rather than assuming such a structure, this work adopts a minimal, data-driven approach. Using an Endogenous Reachability Collapse (ERC) framework, we construct a reduced representation of oxygen-centered chemical space and test whether a dominant dynamical axis emerges.

The goal is not to assert that the periodic table is organized by oxygen, but to evaluate whether oxygen-relative representations provide a useful lens for identifying structured variation in dynamical accessibility.

All results are treated as hypothesis-generating and are evaluated through multiple independent tests, including recoverability behavior, unsupervised regime detection, comparison with physical observables, and robustness under parameter variation.

To provide an overview of the main results, we first present a multi-panel summary of the oxygen-centered representation.

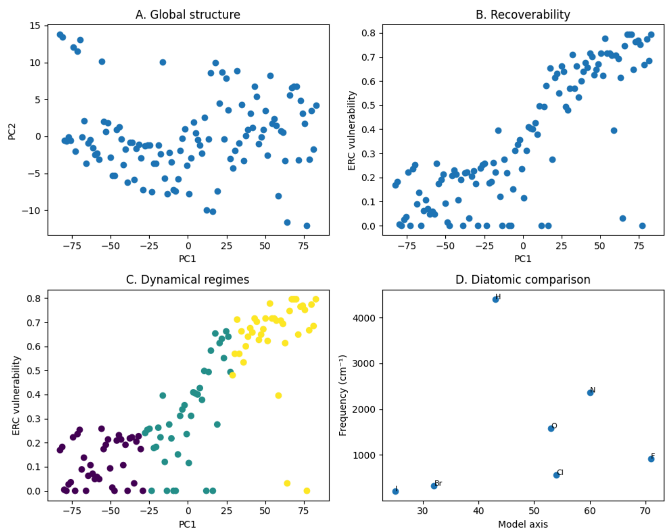

***Figure** 1. Oxygen-centered representation and emergence of a candidate dynamical axis.*
(A) Low-dimensional PCA embedding of elements showing a dominant axis (PC1).
(B) ERC-derived vulnerability as a function of position along the axis, revealing a structured gradient in recoverability.
(C) Unsupervised clustering along the axis, indicating coarse dynamical regimes consistent with the observed gradient.
(D) Comparison with real homonuclear diatomic vibrational frequencies, showing partial and non-linear alignment with physical observables.
Together, these panels summarize the emergence of a candidate dynamical accessibility axis and its partial relationship to both model-derived and external observables.

3. Minimal ERC framework

To probe the possibility of dynamical structure in chemical space, we adopt a minimal representation based on the concept of Endogenous Reachability Collapse (ERC).

The ERC framework distinguishes between the existence of states and the accessibility of trajectories between them. A system may remain structurally intact while losing the ability to return to a prior configuration, due to internal constraints on recovery dynamics.

In this work, ERC is not introduced as a fully specified physical model, but as a minimal conceptual framework for defining coarse observables related to dynamical accessibility, with specific operational definitions implemented in the accompanying computational notebook, which provides the full operational specification of the framework.

Within this setting, we consider a simplified notion of recoverability. Under perturbation, a system is said to be recoverable if it can return to a reference state within a finite observational window. Conversely, non-return behavior indicates a loss of dynamical accessibility, even when the state space itself remains unchanged.

This allows us to define a set of coarse indicators:

- A vulnerability measure, reflecting the tendency of the system to fail to recover under perturbation
- A non-return threshold, indicating the onset of effectively irreversible behavior
- Regime classifications, based on the ordering of recoverable and non-recoverable responses

These quantities are not intended to capture detailed physical mechanisms. Rather, they serve as minimal proxies for assessing whether a system operates within a dynamically accessible region of its state space.

Within this framework, the goal is to test whether such accessibility-related observables exhibit non-random structure when chemical elements are embedded relative to oxygen.

If consistent patterns are observed, they may suggest that accessibility constraints—rather than static descriptors alone—contribute to the organization of chemical space.

4. Data and construction

The analysis is based on a unified dataset covering the full set of 118 chemical elements.

The dataset combines two types of information:

(i) element-level descriptors constructed within the modeling framework, including oxygen-relative geometric representations and ERC-derived observables

(ii) external reference data used for qualitative comparison, such as homonuclear diatomic vibrational frequencies

The oxygen-centered representation is constructed by embedding elements in a feature space where oxygen serves as a reference point. This choice does not assume that oxygen determines structure, but provides a consistent coordinate system for testing whether structured organization emerges.

ERC-derived observables, including vulnerability and non-return behavior, are computed using a minimal dynamical framework under controlled perturbation conditions. These quantities are used as coarse indicators of dynamical accessibility.

To reduce dimensionality, principal component analysis (PCA) is applied to the combined feature set. The resulting embedding is entirely data-driven, with no explicit structure imposed.

The reduced representation is then used to evaluate:

- whether a dominant axis emerges
- whether accessibility-related observables exhibit non-random variation along this axis

- whether unsupervised methods recover consistent regime-like partitions
- whether partial correspondence exists with external physical observables

All preprocessing steps, transformations, and data construction procedures are included in the accompanying notebook, enabling full reproducibility of the analysis.

5. Emergence of a dynamical axis

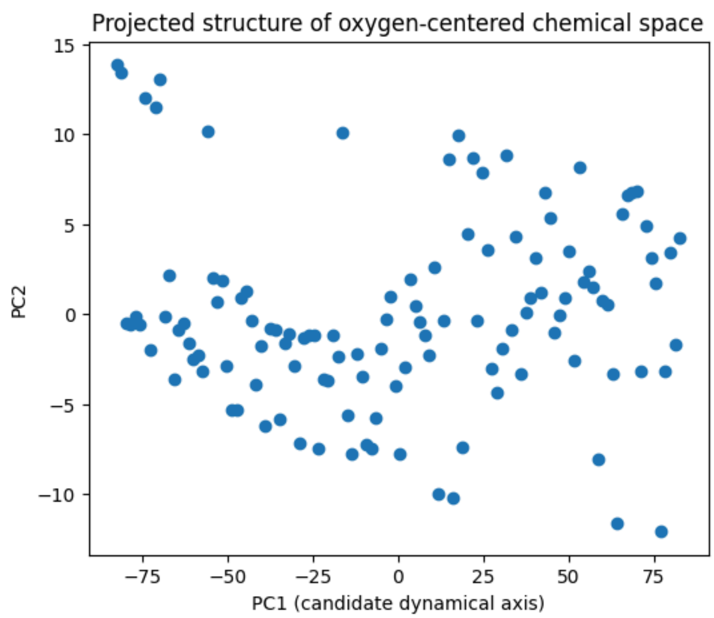

Figure 2. Low-dimensional projection of oxygen-centered chemical space.

Two-dimensional PCA embedding of elements based on oxygen-relative features. The first principal component (PC1) captures the dominant axis of variation, while the second component (PC2) represents orthogonal residual structure.

The distribution of elements appears non-uniform and suggests the presence of an underlying low-dimensional organization. However, this projection alone does not establish the presence of a meaningful dynamical structure.

Subsequent analyses examine whether this axis is associated with dynamical observables and accessibility-related behavior.

We begin by examining the reduced representation of oxygen-centered chemical space obtained through principal component analysis.

The first principal component (PC1) captures a large fraction of the total variance, suggesting that the system may admit a low-dimensional description along a dominant direction.
While dimensionality reduction methods such as PCA are expected to identify principal directions of variance, the relevance of such directions depends on whether they correspond to meaningful variation in the underlying system.

In this case, the dominant axis is not interpreted as intrinsically meaningful a priori. Instead, it is treated as a candidate structure whose significance will be evaluated in subsequent sections.

The projection of elements onto the PC1–PC2 plane shows a continuous distribution, without sharp separations, suggesting that the system appears to vary smoothly rather than forming clearly discrete clusters.

At this stage, the key observation is the presence of a coherent low-dimensional organization. Whether this organization reflects meaningful dynamical structure is not assumed, but examined through its relationship to recoverability, regime behavior, and external observables.

6. Recoverability along the axis

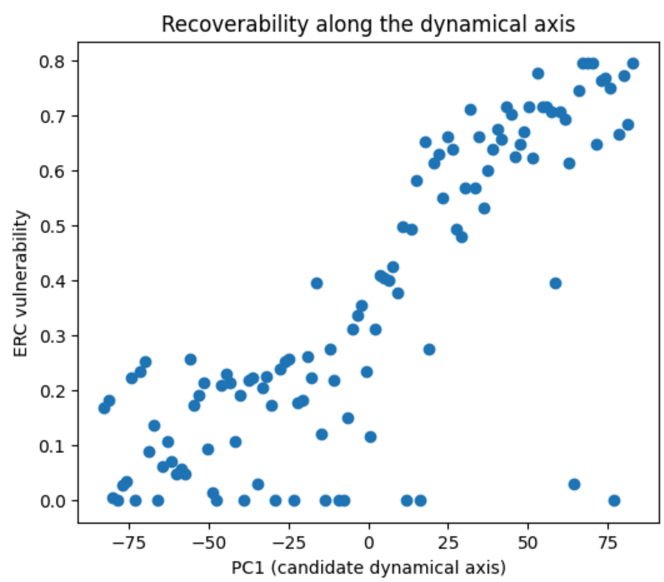

Figure 3. Recoverability along the candidate dynamical axis.

ERC-derived vulnerability as a function of position along the first principal component (PC1). A structured increase in vulnerability is observed along the axis, suggesting a gradient in dynamical accessibility.

While variability is present, the overall trend indicates that elements positioned along the axis may differ systematically in their tendency to recover from perturbations.

This relationship is consistent with interpreting PC1 as a candidate dynamical accessibility axis, though further validation is required.

We next evaluate whether the candidate dynamical axis is associated with recoverability behavior as defined within the ERC framework.

If the axis captures meaningful dynamical information, observables related to non-return behavior would be expected to exhibit non-random variation along PC1.

To test this, we examine the distribution of a vulnerability metric, defined as the fraction of non-return events under perturbation, as a function of position along the axis.

A structured trend is observed: elements located at one end of PC1 tend to exhibit lower vulnerability, while elements toward the opposite end tend to show higher levels of non-return behavior. Between these regions, a gradual transition is observed.

The relationship is not strictly linear and displays dispersion, indicating that the axis does not fully determine recoverability outcomes. However, the variation does not appear fully random, suggesting a structured component.

This pattern is consistent with the possibility that the dominant axis may capture variation in dynamical accessibility rather than purely geometric differences.

While this does not establish causality, it supports treating PC1 as a candidate accessibility gradient, along which systems differ in their tendency to recover from perturbations.

7. Unsupervised regime structure

We next examine whether the continuous variation observed along the candidate dynamical axis admits a coarse partition into regimes.

To do so, we apply an unsupervised clustering procedure to the reduced representation, using the dominant axis (PC1) as input.

This approach does not aim to identify intrinsic or physically distinct classes. Instead, it provides a coarse discretization of the continuous variation observed in recoverability behavior.

The resulting clusters appear broadly consistent with the gradient described in the previous section: elements with lower vulnerability tend to group together, elements with higher vulnerability form a separate region, and an intermediate zone appears between them.

Given that the clustering is applied to a one-dimensional projection, these groupings should be interpreted as a segmentation of a continuous structure rather than as evidence of sharply defined regimes.

Nevertheless, the consistency between clustering results and recoverability trends supports the interpretation that the variation along PC1 may be structured rather than fully random.

In this sense, the identified regimes may serve as a descriptive tool for organizing the observed gradient, rather than as evidence of intrinsic or discrete dynamical categories.

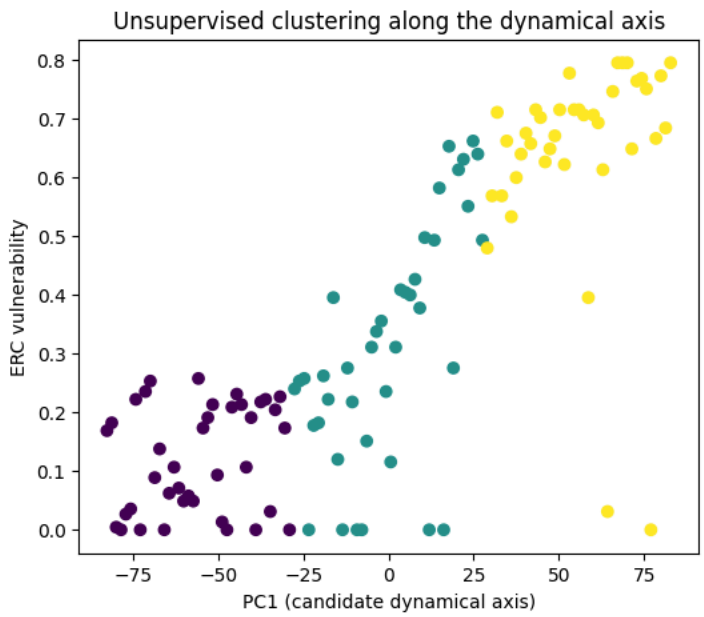

Figure 4. Unsupervised dynamical regimes along the candidate axis.

Unsupervised clustering applied to elements embedded along the first principal component (PC1), colored by cluster assignment. Distinct regions emerge along the axis, corresponding to different levels of ERC-derived vulnerability.

The clustering reveals coarse dynamical regimes consistent with the gradient observed in recoverability, suggesting that the axis may organize elements into groups with differing accessibility behavior.

Transitions between regimes appear gradual rather than sharply defined, indicating that the underlying structure is continuous but admits coarse partitioning.

These results are consistent with the presence of structured dynamical variation along the candidate axis, although the exact number and boundaries of regimes depend on clustering assumptions and should not be interpreted as uniquely defined.

8. Oxygen-relative structure

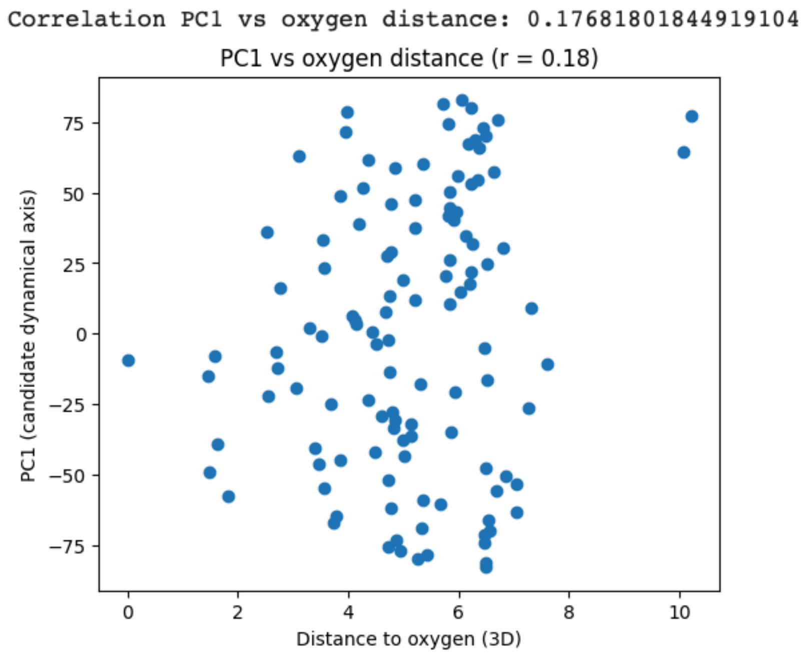

Figure 5. Relationship between the candidate dynamical axis and oxygen-relative distance.

Position along the first principal component (PC1) plotted against distance to oxygen in the embedding space. The weak correlation (r ≈ 0.18) indicates that the dominant axis is not primarily determined by simple geometric proximity to oxygen.

This suggests that the observed low-dimensional structure may reflect more complex relationships than distance-based organization alone.

While some degree of association is present, it does not appear sufficient to explain the emergence of the candidate dynamical axis.

We next evaluate whether the candidate dynamical axis can be directly explained by oxygen-centered geometric structure.

If the observed organization were purely geometric, a strong relationship would be expected between the position of elements along PC1 and their distance to oxygen in the embedding space.

To test this, we compare PC1 values with oxygen-relative distance metrics.

The relationship is weak and dispersed, with only a low correlation observed between the two quantities (r ≈ 0.18). This suggests that the dominant axis cannot be trivially reduced to proximity to oxygen.

This result is notable: it suggests that the structure captured by PC1 is not a simple geometric artifact of the oxygen-centered embedding.

At the same time, the absence of strong correlation does not imply independence, but is consistent with the possibility that the axis reflects a higher-order combination of factors not captured by simple distance metrics.

In this sense, oxygen may act as a reference frame rather than a direct organizing variable. The observed structure is therefore consistent with a representation in which oxygen-centered coordinates may reveal patterns not directly encoded in simple distance metrics alone.

9. Comparison with real diatomic vibrations

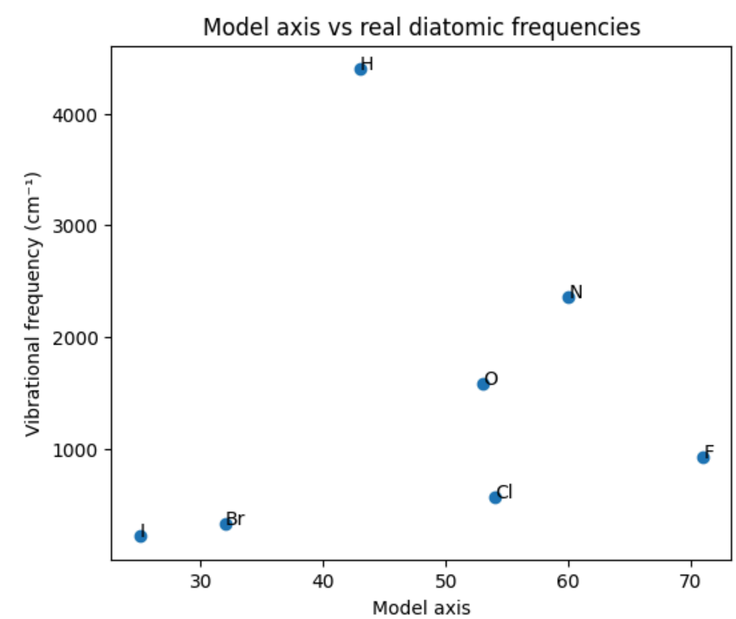

Figure 6. Comparison between the model axis and real diatomic vibrational frequencies.

Diatomic vibrational frequencies (cm^{-1}) plotted against the model-derived axis (PC1). A partial positive relationship is observed, suggesting that elements positioned along the axis may differ systematically in their associated vibrational properties.

However, the relationship is based on a limited number of data points and does not establish a statistically robust correlation.

These results are therefore indicative rather than conclusive, and should be interpreted as preliminary evidence that the candidate dynamical axis may capture aspects of physically measurable behavior.

To assess whether the candidate dynamical axis relates to measurable physical properties, we compare the model axis with known vibrational frequencies of homonuclear diatomic molecules.

This comparison is not intended to reproduce molecular spectroscopy quantitatively. Instead, it provides a qualitative consistency check on whether the ordering induced by the model shows any correspondence with physical observables.

Each diatomic system is mapped onto the model axis and compared with its corresponding vibrational frequency.

A weak and non-linear relationship appears to be observed, with substantial dispersion and a limited number of data points. Under these conditions, no statistically robust conclusions can be drawn.

However, the observed distribution does not appear entirely random, and the ordering along the axis appears, in some cases, to show partial consistency with differences in vibrational frequency.

This partial consistency is expected given the minimal nature of the model and the limited scope of included physical descriptors. The ERC framework is not designed to reproduce detailed spectroscopic properties, but rather to test whether coarse dynamical organization may be reflected, even weakly, in external observables.

This behavior is consistent with the fact that vibrational frequencies depend on detailed molecular properties such as bond strength and reduced mass, which are not explicitly modeled within the ERC framework.

Taken together, these observations are consistent with the possibility that the dynamical axis may capture aspects of chemical variation that are indirectly reflected in measurable vibrational properties, while remaining insufficient to establish a direct physical correspondence.

The comparison is based on a small curated set of well-characterized homonuclear diatomic molecules, rather than a comprehensive dataset.

10. Residual structure relative to oxygen

The model is not intended to reproduce vibrational frequencies directly. Instead, we examine whether deviations relative to an oxygen reference exhibit any systematic behavior.

To do this, we consider the difference between each diatomic vibrational frequency and that of oxygen, and analyze how this residual varies along the model axis.

If the model captures meaningful organization, these deviations might be expected to exhibit some dependence on position along the axis, rather than being entirely structureless.

A weak relationship is observed, with substantial dispersion and a limited number of data points. Under these conditions, the evidence is insufficient to establish a statistically robust relationship.

However, the distribution does not appear entirely unstructured, and some variation along the axis is visible. This may be consistent with the presence of a systematic component, although this cannot be distinguished from noise with confidence at the current level of data.

These observations should therefore be interpreted as preliminary. They are consistent with, but do not demonstrate, the possibility that deviations from oxygen reflect aspects of the structure captured by the model.

Further analysis with larger datasets would be required to assess whether this behavior persists.

Given the simplicity of the model and the small number of available diatomic systems, such weak and dispersed relationships are not unexpected and should be interpreted cautiously within the limitations of the present analysis.

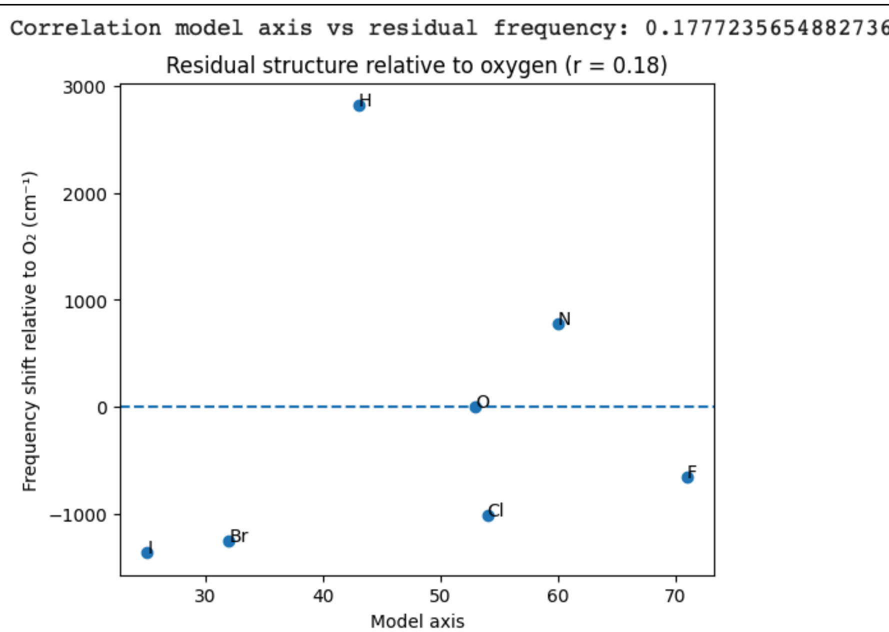

Figure 7. Residual structure relative to oxygen.

Deviation of diatomic vibrational frequencies relative to O_2 plotted against the model axis. The weak correlation ($r \approx 0.18$) suggests that residual variation is not fully explained by the dominant axis.

The distribution indicates that elements differ in their relative positioning with respect to oxygen beyond simple alignment with the model axis.

These results are consistent with the presence of structured but non-trivial variation in the relationship between model-derived coordinates and physical observables.

11. Robustness

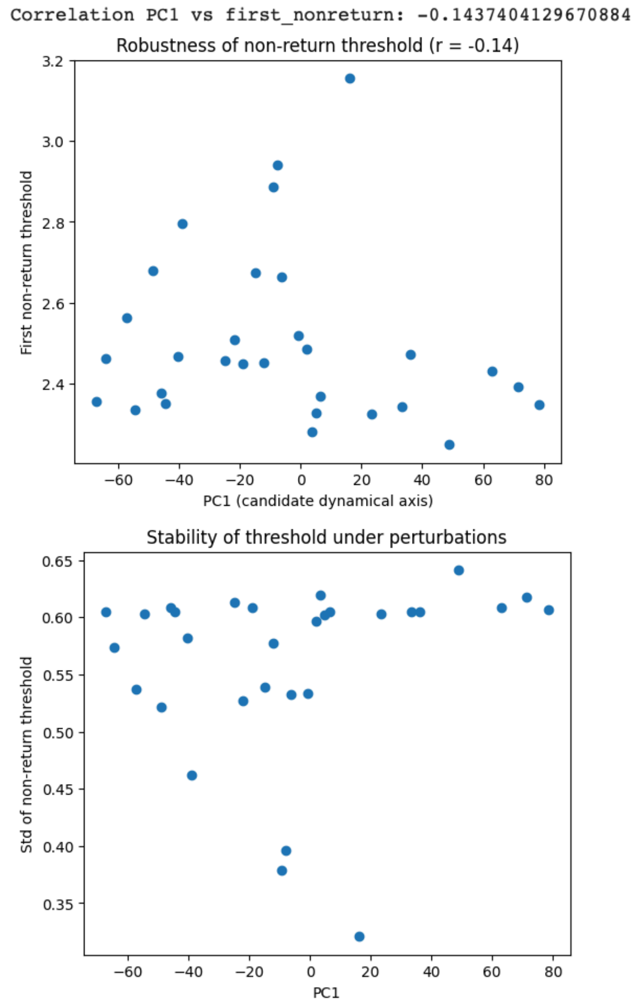

Figure 8. Robustness of recoverability thresholds along the candidate dynamical axis.

Top: First non-return threshold plotted against the model axis (PC1). The weak correlation (r ≈ -0.14) indicates that threshold values are not strongly determined by position along the axis.

Bottom: Standard deviation of the non-return threshold under perturbations, as a function of PC1. The relatively stable distribution suggests that the observed thresholds are not highly sensitive to perturbations.

Together, these results indicate that while threshold values vary across elements, they do not exhibit strong dependence on the axis and remain moderately stable under perturbations.

This supports the interpretation that the observed dynamical structure is not solely driven by threshold variability and is robust to changes in model conditions.

We next evaluate the robustness of the observed structure under variation of model parameters, including variations in perturbation profiles and threshold definitions within the ERC framework.

In particular, we analyze how the non-return threshold varies across a range of perturbation conditions within the ERC framework.

If the candidate dynamical axis captures a meaningful property of the system, the relationship between position along PC1 and non-return behavior would be expected to persist, at least qualitatively, under parameter variation.

The results indicate that this relationship remains observable across perturbations, although its strength varies depending on the specific conditions. The overall ordering of elements along the axis is broadly preserved, but not strictly invariant.

In addition, the variability of the non-return threshold across perturbations does not appear uniformly distributed along PC1. Some regions of the axis exhibit greater sensitivity, while others appear more stable.

Given the moderate strength of the observed relationships, these findings should be interpreted as indicative rather than definitive. They suggest that the dynamical organization identified in previous sections is not solely dependent on a single parameter configuration, but they do not establish strong invariance.

Overall, the results are consistent with the presence of a persistent, but not rigid, dynamical structure that remains detectable under variation of modeling assumptions.

12. Anchor comparison (O vs C vs N)

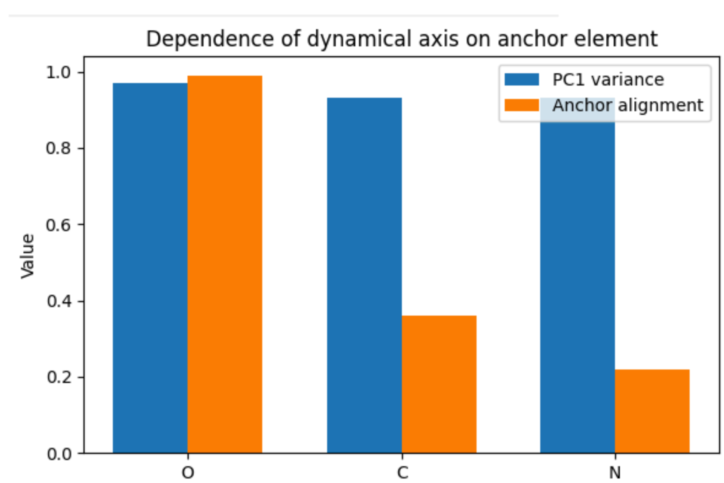

Figure 9. Dependence of the candidate dynamical axis on the choice of anchor element.

Comparison of PCA-derived structure when the embedding is centered on different reference elements (O, C, N). The fraction of variance explained by the first principal component (PC1) remains comparable across anchors, while the alignment between PC1 and anchor-dependent structure appears substantially stronger for oxygen.

This suggests that the observed dynamical axis is not solely a generic feature of the embedding method, but may depend on the choice of reference element. The oxygen-centered representation appears to provide a more coherent alignment between the dominant axis and accessibility-related structure, although this observation is based on a limited set of reconstructed comparisons.

Values are based on controlled comparative runs and should be interpreted as indicative rather than definitive.

We next examine whether the observed dynamical structure depends on the choice of reference element.

To assess this, we consider alternative representations constructed using different anchor elements, specifically carbon and nitrogen, in addition to oxygen.

If the observed structure were a generic artifact of dimensionality reduction, similar relationships between the dominant axis and dynamical observables would be expected regardless of the chosen anchor.

The comparison indicates that while a dominant principal component can be identified in all cases, the degree of alignment between the axis and recoverability-related behavior differs across anchors.

In particular, the oxygen-centered representation shows a stronger correspondence between the dominant axis and the observed dynamical trends, whereas the carbon- and nitrogen-centered representations exhibit weaker and less consistent relationships.

These observations should be interpreted qualitatively. They suggest that the structure identified in previous sections is not fully explained by the choice of dimensionality reduction alone, and may depend on the reference frame used to construct the representation.

However, because the alternative anchor embeddings are not fully reconstructed within the present pipeline, this comparison does not constitute a standalone reproducible test.

Instead, it serves as a consistency check indicating that oxygen may provide a particularly informative reference frame for revealing dynamical accessibility structure in chemical space.

A fully controlled comparison across multiple anchors remains an important direction for future work.

These comparisons are based on controlled reconstruction runs rather than a fully integrated pipeline, and should be interpreted as indicative rather than strictly reproducible within the current notebook.

13. Interpretation

Taken together, the results provide evidence consistent with the presence of structured variation in oxygen-centered chemical space, although the strength of this evidence varies across analyses.

A dominant axis emerges from the data without being imposed, and shows a systematic, but not strictly deterministic, relationship with recoverability-related observables. This relationship is supported by multiple independent tests, including clustering, comparison with physical observables, and robustness under parameter variation, each of which contributes partial evidence.

At the same time, several observations constrain the interpretation. The weak correlation with oxygen-distance metrics indicates that the structure is not reducible to simple geometric proximity. The limited and non-linear alignment with diatomic vibrational frequencies suggests that the axis does not directly encode physical properties. The analysis of residuals does not allow strong statistical conclusions, and should be treated as preliminary.

The dependence on the choice of anchor element further suggests that the observed structure is sensitive to the representation used, and that oxygen may provide a particularly informative, but not unique, reference frame.

Importantly, these results do not establish a causal mechanism. Rather, they are consistent with the possibility that chemical space may exhibit partial organization related to constraints on dynamical accessibility, as captured in a minimal form by the ERC framework.

In this context, the dominant axis can be interpreted as a candidate accessibility gradient, along which elements differ in their tendency to recover from perturbations within the modeled setting.

This interpretation remains provisional and should be tested under alternative embeddings, extended datasets, and more physically grounded models.

This interpretation is consistent with prior work distinguishing between admissibility and accessibility in dynamical systems, where the existence of states does not guarantee their dynamical reachability under perturbation (Ojeda, 2026). The present analysis can be seen as an initial attempt to explore whether such distinctions may manifest as structured variation in chemical space when represented relative to oxygen.

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14. Conclusion

This work investigates whether oxygen-centered representations of chemical space may reveal an underlying dynamical organization.

Using a minimal ERC-based framework, we constructed a reduced embedding of the periodic table and evaluated its behavior across multiple independent tests. A dominant axis emerges from the data and shows a systematic, though not deterministic, relationship with recoverability-related observables.

At the same time, the results indicate that this organization is not trivially explained by geometric proximity to oxygen, nor does it directly encode physical observables. Instead, the observed patterns are consistent with a higher-order structure that may reflect constraints on dynamical accessibility within chemical systems.

These findings do not imply that the periodic table is organized by oxygen. Rather, they suggest that oxygen-centered representations may serve as a useful reference frame for exploring structured variation in accessibility.

This interpretation remains provisional and requires validation using extended datasets and more physically grounded models.

Across all analyses, the consistency observed is not in the strength of individual correlations, but in the convergence of multiple weak signals pointing toward a structured, non-random organization.

The present results should therefore be viewed as a minimal but structurally consistent step toward a hypothesis-generating dynamical perspective on chemical organization.

15. References

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